

Notes: Joh (JFE, 2003)

Corporate Governance and Firm Profitability: Evidence from Korea Before the Economic Crisis

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1 Big picture

- **Research question.** Did weak corporate governance and controlling-shareholder incentives reduce Korean firm profitability before the 1997 economic crisis?
- **Main idea.** When controlling shareholders have control rights that exceed their cash-flow ownership rights, they bear only part of the cost of resource diversion but receive the private benefits. This wedge can lower firm profitability.
- **Empirical setting.** Korea is a useful setting because, before the crisis, it combined weak minority-shareholder protection, protected incumbent controlling shareholders, high leverage, low profitability, and extensive business groups, or *chaebols*.
- **Empirical object.** The paper studies 5,829 Korean firms subject to outside auditing during 1993–1997, producing 19,497 firm-year observations. Profitability is mainly measured by ordinary income/assets and net income/assets.
- **Main finding.** Firms with lower controlling-family ownership have lower profitability. Firms with larger control–ownership disparities also have lower profitability, especially among publicly traded firms and business-group firms.
- **Business-group result.** Independent firms outperform firms affiliated with large *chaebols*. The evidence is inconsistent with the idea that large business groups were still mainly value-enhancing internal markets in Korea during the mid-1990s.
- **Tunneling result.** Investments in affiliated firms reduce profitability, while investments in unaffiliated firms tend to increase profitability. This supports the view that internal capital markets were used to transfer or waste resources within groups.
- **Important distinction.** The paper is about pre-crisis operating profitability, not only crisis-period stock-market losses. Poor governance appears to have weakened firms before the crisis became visible.
- **Economic takeaway.** Weak governance can create macro-financial fragility through micro-level profitability losses: if firms remain overlevered and unprofitable for years, the banking system becomes vulnerable to nonperforming loans.

2 Incremental contribution of the paper

- **From crisis outcomes to pre-crisis causes.** Earlier papers emphasize governance during the Asian financial crisis. Joh studies whether poor governance reduced profitability before the crisis.
- **Firm-level evidence inside one country.** The paper avoids many cross-country comparability problems by studying firms operating under the same legal system, accounting standards, macroeconomic conditions, and institutional environment.
- **Private and public firms together.** The sample includes privately held firms as well as listed firms. This matters because private firms make up most of the observations and because stock-market measures cannot be used for them.

- **Accounting profitability rather than market valuation.** The paper uses profitability measures that are directly related to financial survivability. This is especially important in an emerging market where stock prices may not fully reflect information.
- **Continuous ownership and wedge measures.** Instead of relying only on a dummy for high control rights, the paper measures controlling-family ownership and the control–ownership wedge continuously.
- **Direct evidence on tunneling.** The paper distinguishes investments in affiliated firms from investments in unaffiliated firms, allowing a direct test of whether group internal capital markets destroy firm profitability.
- **Chaebol evidence.** The paper shows that affiliation with large business groups is negatively associated with profitability, challenging the view that group affiliation is necessarily beneficial in emerging markets.
- **Public-firm amplification.** The governance effects are often stronger among publicly traded firms, where controlling shareholders typically own smaller cash-flow stakes but retain effective control.

3 Data and measurement

3.1 Firm sample and data sources

The data come from the National Information and Credit Evaluation database. Firms submit financial statements to the Korea Securities Supervisory Board, and NICE compiles and checks the data.

- **Sample.** 5,829 Korean firms during 1993–1997.
- **Observations.** 19,497 firm-year observations.
- **Coverage rule.** Firms are subject to outside auditing and had assets exceeding 6 billion won in 1997.
- **Exclusions.** Financial institutions and state-controlled firms are excluded.
- **Listing status.** About 24% of observations are publicly traded and 76% are privately held.
- **Business groups.** About 27% of observations are group-affiliated, and about 12% belong to the largest 70 *chaebols*.

Interpretation. The dataset is broad enough to compare governance effects across listed and unlisted firms, independent firms and business-group firms, and firms with very different ownership structures.

3.2 Profitability measures

The main dependent variables are accounting profitability ratios:

- ordinary income divided by assets;

- net income divided by assets;
- ordinary income divided by sales and net income divided by sales in robustness tests.

Ordinary income is operating income minus interest payments plus dividends, gains on securities, and related items. Net income subtracts taxes and extraordinary items from ordinary income.

Why accounting profitability? Joh argues that accounting profitability is useful because stock-market prices may be inefficient in emerging markets, accounting measures better capture financial survivability, and accounting data can be used for privately held firms.

3.3 Ownership and control variables

- **Ownership concentration.** The direct ownership stake of the largest individual shareholder and family members.
- **Control–ownership disparity.** The difference between control rights and ownership rights, similar to the control-rights wedge used in the corporate-governance literature.
- **Large 70 chaebol dummy.** Equals one if the firm belongs to one of the 70 largest business groups.
- **Large 30 chaebol dummy.** Used in robustness tests as a narrower business-group definition.

Interpretation. High ownership concentration aligns the controlling family with other shareholders. A large control–ownership wedge means the controlling family can influence resources while bearing a smaller fraction of losses.

3.4 Other firm characteristics

The regressions also include controls for size, capital structure, market position, and business strategy:

- log assets or log sales;
- lagged equity ratio;
- market share in the four-digit industry;
- exports/sales;
- advertising/sales;
- investments in affiliated firms/assets;
- investments in unaffiliated firms/assets;
- public-firm dummy and its interactions in the tunneling tests.

4 Methodology and empirical specifications

4.1 Baseline profitability specification

The core regression relates firm profitability to ownership and firm characteristics while controlling for industry and time effects:

$$Profitability_{ijt} = \alpha_j + \tau_t + \beta Ownership_{it} + \delta Chaebol_i + \Gamma' X_{it} + \varepsilon_{ijt},$$

where i indexes firms, j indexes four-digit industries, and t indexes years. The paper estimates within-unit specifications, so the coefficient measures how a firm's deviation from the industry mean relates to its profitability deviation from the industry mean.

- α_j controls for unobserved industry fixed effects.
- τ_t controls for macroeconomic time effects.
- X_{it} contains size, capital structure, market share, export intensity, advertising intensity, and other firm characteristics.

4.2 Ownership concentration versus control–ownership wedge

The paper estimates two main versions of the governance regression:

$$Profitability_{ijt} = \alpha_j + \tau_t + \beta_1 Ownership_{it} + \delta Chaebol_i + \Gamma' X_{it} + \varepsilon_{ijt},$$

and

$$Profitability_{ijt} = \alpha_j + \tau_t + \beta_2 Wedge_{it} + \delta Chaebol_i + \Gamma' X_{it} + \varepsilon_{ijt}.$$

A positive β_1 means higher controlling-family ownership is associated with higher profitability. A negative β_2 means a larger control–ownership disparity is associated with lower profitability.

4.3 Firm fixed effects and industry fixed effects

The paper first shows that ownership concentration predicts profitability even with firm fixed effects. It then focuses on industry fixed effects because business-group affiliation does not vary enough within firms to estimate its effect under firm fixed effects.

Interpretation. Firm fixed effects help address time-invariant unobserved firm heterogeneity. Industry fixed effects allow the paper to estimate the effect of chaebol affiliation and compare firms within narrowly defined industries.

4.4 Yearly regressions

Joh estimates separate regressions by year to see whether ownership effects are driven by one year or by the crisis year only.

- Ownership concentration is positive in most years.
- The ownership effect is particularly strong in 1997.
- Large *chaebol* affiliation is negative in every year.

4.5 Subsample tests

The paper estimates the main regressions on three samples:

- all sample firms;
- publicly traded firms only;
- business-group-affiliated firms only.

Interpretation. These subsamples test whether the governance problem is stronger when minority shareholders are more exposed: listed firms have dispersed minority investors, and group-affiliated firms have more tools for tunneling.

4.6 Time-varying ownership effects

To study whether expropriation is worse in downturns, the paper estimates:

$$\begin{aligned} Profitability_{ijt} = & \alpha_j + \tau_t + \beta Ownership_{it} \\ & + \theta (Ownership_{it} \times GNPgrowth_t) \\ & + \delta Chaebol_i + \Gamma' X_{it} + \varepsilon_{ijt}. \end{aligned}$$

A negative θ means ownership matters more when aggregate growth is lower, consistent with controlling shareholders extracting more when financial distress is more likely.

4.7 Nonlinear ownership effects

The paper tests whether ownership effects are nonlinear in two ways:

- piecewise linear splines over 0–5%, 5–25%, and above 25%;
- a cubic function of ownership concentration.

Interpretation. This allows for both incentive-alignment and entrenchment effects. The Korean evidence suggests that entrenchment can appear even at very low ownership levels because controlling shareholders are protected from outside discipline.

4.8 Internal capital markets and tunneling

The paper tests whether investments in affiliated firms reduce profitability:

$$\begin{aligned}
Profitability_{ijt} = & \alpha_j + \tau_t + \beta Ownership_{it} \\
& + \theta_1 InvAffiliated_{it} + \theta_2 InvUnaffiliated_{it} \\
& + \delta Chaebol_i + \Gamma' X_{it} + \varepsilon_{ijt}.
\end{aligned}$$

The paper also interacts investment variables with a public-firm dummy to test whether tunneling is stronger in publicly traded firms.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies whether firms with governance structures that create stronger controlling-shareholder incentives have lower accounting profitability within Korea before the crisis. The evidence is strongest as a firm-level association supported by fixed effects, controls, subsample tests, and robustness checks.

5.2 Why Korea is a useful setting

Studying one country avoids some cross-country endogeneity problems. Korean firms in the sample face the same broad legal environment, accounting standards, macroeconomic conditions, and institutional constraints. This makes ownership and business-group variation more interpretable than in a broad cross-country sample.

5.3 Why the ownership variables matter

Low controlling-family ownership means the family bears a smaller fraction of the cost of inefficient investment or diversion. A high control-ownership wedge means the family can still exercise control while holding limited cash-flow rights. Both structures create incentives for expropriation.

5.4 Why the public-versus-private comparison matters

Publicly traded firms have more outside minority shareholders, while controlling families often hold lower ownership stakes. If tunneling is more severe in public firms, the interpretation is consistent with minority-shareholder expropriation rather than only poor management quality.

5.5 Endogeneity concern

Ownership structure may itself respond to profitability or unobserved firm quality. Firm fixed effects and industry fixed effects reduce this concern but do not eliminate it. The paper is transparent that the investment-in-affiliated-firms tests can have endogeneity issues because firms choose those investments.

5.6 Measurement concern

Accounting profitability can be affected by reporting quality. Joh argues that using firms subject to outside auditing, examining several accounting profitability measures, and studying one accounting regime helps reduce this concern.

5.7 Interpretation of the main coefficients

The ownership coefficient should be read as a reduced-form governance association: firms with higher controlling-family ownership have higher profitability, consistent with better incentive alignment. The wedge coefficient should be read as evidence that separating control from cash-flow rights is costly for operating profitability.

6 Tables and figures

6.1 Figure 1: Profitability of Korean firms from 1967 to 1997

Read:

- Figure 1 plots ordinary income on equity against the average borrowing interest rate for Korean firms from 1967 to 1997.
- Ordinary income on equity is often below the borrowing interest rate in the pre-crisis period.
- Profitability collapses in 1997, but the figure also shows that low profitability was a persistent issue before the crisis.

Economic message: Korea's corporate sector was fragile before the crisis. High leverage is especially dangerous when firms cannot consistently earn returns above their borrowing costs.

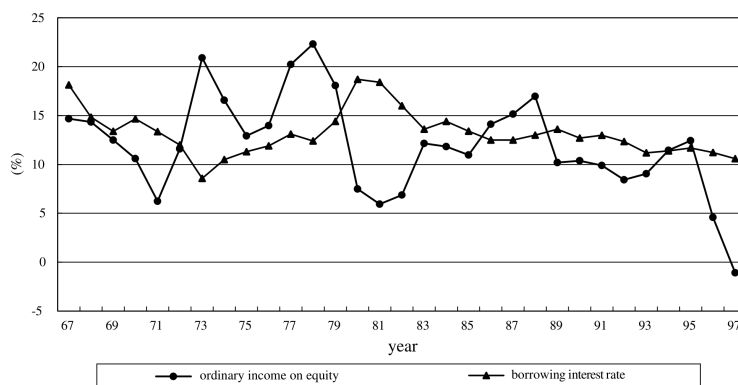


Figure 1: Profitability of Korean firms from 1967 to 1997.

6.2 Tables 1 and 2: Ownership structure and summary statistics

Read Table 1:

- The average controlling-family ownership stake is 45.19% for all firms, but the asset-weighted mean is only 19.65%.
- Publicly traded firms have much lower family ownership: 31.67% on average and 13.05% asset-weighted.
- Large 70 *chaebol*-affiliated firms have especially low ownership: 17.12% on average and 9.91% asset-weighted.
- The average control–ownership disparity is 20.97% for all firms and 43.52% for large 70 *chaebol* firms.
- By 1997, the control–ownership disparity for large 70 *chaebol* firms rises to 49.63%.

Read Table 2:

- Mean net income/assets is only 0.2478%, and mean ordinary income/assets is 1.2372%.
- Profitability deteriorates over time: net income/assets falls from 1.0096% in 1993 to -1.7290% in 1997.
- Firms invest 9.54% of assets in unaffiliated firms and 3.26% of assets in affiliated firms, on average.
- The sample is 24.28% publicly traded firms, 27.25% business-group-affiliated firms, and 12.23% large 70 *chaebol* firms.

Economic message: Korean controlling shareholders often retained control with limited cash-flow ownership, especially in public and *chaebol* firms. This is precisely the setting where minority-shareholder expropriation should be most severe.

Table 1
Controlling shareholders' ownership stakes
Using the ownership information in the NCFE data set, this table shows the largest individual shareholder's and family members' ownership and differences of control rights and ownership rights (control-ownership) of 5,829 firms during 1993–1997. All firms were subject to outside auditing and had assets exceeding 6 billion won in 1997. Of the total sample of 19,497 observations, 24% are publicly traded and 76% are privately held. Independent firms account for 73%, and group affiliated firms account for 27% of total observations. Top 70 large business group affiliated firms (*chaebol*) account for 12%. Panel A reports the simple mean and size-adjusted mean per firm over all years when using firm asset as weight. Panel B reports the average per firm each year.

Panel A:

Type of firms	Ownership concentration					Difference between control rights and ownership rights		
	Num. of obs.	Mean (%)	Standard deviation	Weighted Mean (%)	Mean (%)	Standard deviation	Weighted Mean (%)	
All firms	19,497	45.19	34.61	19.65	20.97	33.37	22.25	
Publicly traded firms	4,702	31.67	28.10	13.05	15.22	25.54	17.89	
Privately held firms	14,796	49.49	35.37	38.02	22.80	35.50	34.36	
Independent firms	14,184	51.19	33.35	41.83	19.23	30.74	16.67	
Group affiliated firms	5,314	29.20	32.79	12.55	32.45	37.17	23.22	
Large <i>chaebol</i> affiliated firms	2,385	17.12	26.97	9.91	43.52	38.49	24.16	

Panel B:

Type of firms	Ownership					Difference between control rights and ownership rights				
	1993	1994	1995	1996	1997	1993	1994	1995	1996	1997
All firms	43.36	43.65	46.01	46.72	46.32	18.69	20.87	20.86	21.55	23.47
Publicly traded firms	33.48	31.74	32.78	31.31	26.57	13.69	15.02	15.28	15.95	17.33
Privately held firms	49.58	48.02	50.07	50.86	51.10	20.69	23.02	22.58	23.06	24.96
Independent firms	47.33	49.87	51.63	52.34	52.45	14.38	16.35	16.84	17.26	18.73
Group affiliated firms	29.17	29.22	31.31	30.82	21.75	28.52	31.38	31.40	33.69	42.45
Large 70 <i>chaebol</i> affiliated firms	17.23	17.32	18.64	17.26	15.37	38.09	42.12	43.87	38.31	49.63

Table 1: Controlling shareholders' ownership stakes.

Table 2
Summary statistics
The summary statistics show the time-series average of the cross-sectional statistics for 5,829 firms during 1993–1997 (total 19,497 observations). All firms were subject to outside auditing and had assets exceeding 6 billion won in 1997. Ownership stake is the sum of the largest individual shareholder's and family's ownership. The "Public firm" dummy has a value of 1 when the firm is listed in either the Korean Stock Exchange or the KOSDAQ (Korea Securities Dealers Automated Quotation).

Variables	All year		1993		1994	1995	1996	1997
	Mean	Std	Mean	Mean	Mean	Mean	Mean	Mean
Net income to assets (%)	0.2478	10.6112	1.0096	1.5540	0.4427	−0.5027	−1.7290	
Ordinary income to assets (%)	1.2372	11.0496	2.0361	2.4466	1.4168	0.4058	−0.5094	
Ownership stake (%)	45.1952	34.6140	43.3562	43.6535	46.0069	46.7204	46.3158	
Difference between control rights and ownership rights (%)	20.9696	33.3652	18.6912	20.8716	20.8688	21.5564	23.4692	
Log (Asset)	3.1782	1.1770	3.1413	3.2148	3.1705	3.1644	3.2103	
Equity ratio (%)	19.4856	32.8244	21.1821	20.7130	19.1823	18.1209	18.0073	
Market share (%)	4.5526	12.2631	4.8129	4.6188	4.5364	4.2550	4.5876	
Export/sales (%)	7.1765	19.7142	8.6293	7.6260	6.8584	6.3458	6.2854	
Advertisement/sales (%)	0.8232	3.2174	0.8244	0.8234	0.8212	0.8404	0.7980	
Investment in unaffiliated firms								
Over assets (%)	9.5445	7.9068	9.2477	9.2984	9.4227	9.8342	10.0214	
Investment in affiliated firms								
Over assets (%)	3.2616	6.4997	2.7192	3.1708	3.5401	3.7283	3.0071	
Business group-affiliated firms	0.2725	0.4453	0.3051	0.3010	0.2767	0.2614	0.1999	
Large 70 <i>chaebol</i> dummy	0.1223	0.3277	0.1381	0.1365	0.1225	0.1147	0.1068	
Publicly traded firm dummy	0.2428	0.4278	0.2869	0.2685	0.2349	0.2118	0.1950	
Number of observations	19,497		3,953	4,007	4,189	4,456	2,892	

Table 2: Summary statistics.

6.3 Tables 3 and 4: Baseline ownership effects and yearly regressions

Read Table 3:

- Ownership concentration is positively related to ordinary income/assets and net income/assets.

- The result appears with firm fixed effects and with industry fixed effects.
- Adding firm size and capital-structure controls does not eliminate the ownership effect.
- The large 70 *chaebol* dummy is strongly negative in the industry-fixed-effect specifications.

Read Table 4:

- The ownership effect is positive in most individual-year regressions.
- The effect is not statistically strong in 1994, a high-growth year.
- The coefficient on ownership concentration is largest in 1997.
- Large 70 *chaebol* affiliation is negative in every year.

Economic message: Better alignment between controlling shareholders and minority shareholders is associated with higher profitability. The relationship becomes stronger when macroeconomic conditions deteriorate.

Table 3
Impact of ownership concentration on firm profitability
The dependent variables are ordinary income to assets and net income to assets. Panel A controls for firm and time-fixed effects, and Panel B controls for industry and time-fixed effects. Each column shows the results of within-unit estimation. All variables are measured as the deviation from the time-series mean of each variable, per firm in Panel A and per industry in Panel B. Large 70 *chaebol* is a dummy variable that takes 1 when a firm belongs to the largest 70 business groups. *T*-statistics are in parentheses.

	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets
Ownership concentration	0.0139 (2.63)	0.0133 (2.35)	0.0166 (3.16)	0.0161 (2.88)	0.0186 (4.14)	0.0180 (3.67)
Log (Assets)			3.9637 (15.15)	4.1161 (16.81)	3.9621 (16.75)	2.1624 (8.80)
Equity ratio					0.2682 (71.50)	0.2622 (63.95)
Firm dummies	Included	Included	Included	Included	Included	Included
Year dummies	Included	Included	Included	Included	Included	Included
No. of observations	19,497	19,497	19,497	19,497	19,497	19,497
R ²	0.7036	0.6365	0.7079	0.6423	0.7874	0.7247

	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets
Ownership concentration	0.0128 (5.44)	0.0095 (4.21)	0.0066 (2.69)	0.0043 (1.81)	0.0148 (5.94)	0.0128 (5.21)	0.0180 (7.27)	0.0153 (6.47)
Large 70 <i>chaebol</i> dummy			-2.3167 (-9.02)	-1.9574 (-7.90)	-3.7279 (-13.57)	-3.4483 (-12.88)	-3.8995 (-11.51)	-2.8973 (-11.09)
Log (Assets)					1.0927 (13.85)	1.1232 (14.78)	0.9429 (12.23)	1.0010 (13.38)
Equity ratio, _{t-1}							0.0729 (31.58)	0.0594 (26.54)
Industry dummies	Included	Included	Included	Included	Included	Included	Included	Included
Year dummies	Included	Included	Included	Included	Included	Included	Included	Included
No. of observations	19,497	19,497	19,497	19,497	19,497	19,497	19,497	19,497
R ²	0.1007	0.0947	0.1043	0.0976	0.1134	0.1077	0.1371	0.1393

Table 3: Impact of ownership concentration on firm profitability.

Table 4
Yearly estimation of the impacts of ownership concentration on firm profitability
The dependent variables are ordinary income to assets and net income to assets. The other explanatory variables are firm characteristics, including the affiliation status to the large 70 business groups, firm size, and financial structure. In each regression, 4-digit industry dummies are included. Large 70 *chaebol* is a dummy variable that has a value of 1 when a firm belongs to the largest 70 business groups. *T*-statistics are in parentheses.

	1993		1994		1995		1996		1997	
	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets
Log (Asset)	0.5002 (2.88)	0.6569 (3.85)	0.4319 (3.12)	0.3942 (2.70)	0.8335 (5.48)	0.8989 (6.20)	1.2885 (8.16)	1.2722 (8.56)	1.2331 (4.99)	1.4114 (6.16)
Equity ratio, _{t-1}	0.0047 (1.06)	0.0034 (0.78)	0.0760 (19.12)	0.0241 (5.75)	0.0921 (20.03)	0.0791 (18.04)	0.1097 (20.22)	0.1028 (20.13)	0.1591 (17.36)	0.1741 (20.52)
Ownership concentration	0.0158 (2.81)	0.0119 (2.15)	0.0071 (1.66)	-0.0001 (-0.03)	0.0118 (3.24)	0.0146 (3.49)	0.0209 (4.14)	0.0173 (3.65)	0.0480 (6.05)	0.0456 (6.20)
Large 70 <i>chaebol</i> dummy	-2.7577 (-4.74)	-3.0489 (-5.33)	-2.0985 (-4.63)	-1.8918 (-3.96)	-2.9028 (-5.49)	-2.0098 (-3.98)	-3.3187 (-5.78)	-3.0696 (-5.66)	-2.3838 (-2.51)	-2.7783 (-3.15)
Industry dummies	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included
No. of observations	3,953	3,953	4,007	4,007	4,189	4,189	4,456	4,456	2,892	2,892
R ²	0.1066	0.1098	0.2293	0.1199	0.2302	0.2137	0.2386	0.2344	0.3066	0.3510

Table 4: Yearly estimation of ownership effects.

6.4 Table 5: Determinants of firm profitability

Read Panel A:

- Ownership concentration remains positive after adding size, equity ratio, export intensity, advertising intensity, market share, year effects, and industry effects.
- The ownership coefficient is especially large for publicly traded firms.
- Large 70 *chaebol* affiliation is negative for all firms and for publicly traded firms.

Read Panel B:

- The control-ownership wedge is negative for publicly traded firms and business-group firms.
- The wedge effect is strongest and most statistically clear among publicly traded firms.
- Equity ratio and market share are generally positively related to profitability, while advertising/sales is generally negative.

Economic message: The core governance result survives a rich set of controls. Low ownership and high control–ownership disparity are not simply proxies for firm size, capital structure, market share, or business strategy.

Table 5
Determinants of firm profitability
This table shows two panels of results on determinants of profitability. The dependent variables are ordinary income to assets and net income to assets. Panel A uses ownership concentration as an explanatory variable. Panel B uses the difference between control rights and ownership rights (control–ownership) as an explanatory variable. The three pairs of columns in each panel show the results using data from (a) all sample firms, (b) only publicly traded firms, and (c) firms affiliated with business groups. Controlling for industry and time-fixed effects, within-unit estimation is used. All variables are measured as the deviation from the time-series industry mean of each variable. Large 70 *chaebol* is a dummy variable that has a value of 1 when a firm belongs to one of the largest 70 business groups. *T*-statistics are in parentheses.

	All sample firms		Publicly traded firms only		Business group affiliated firms	
	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets
Log (Asset)	0.6108 (7.11)	0.7343 (8.81)	0.1321 (0.89)	0.2406 (1.70)	0.6036 (4.26)	0.8164 (5.57)
Equity ratio _{it-1}	0.0723 (31.42)	0.0599 (26.37)	0.0272 (7.41)	0.0235 (6.69)	0.0657 (19.69)	0.0527 (11.59)
Ownership concentration	0.0176 (7.23)	0.0151 (6.39)	0.0506 (9.87)	0.0465 (9.48)	0.0288 (5.40)	0.0213 (3.80)
Large 70 <i>chaebol</i> Dummy	−3.2539 (−12.11)	−3.0300 (−11.62)	−1.3560 (−3.54)	−1.2592 (−3.43)	−0.4191 (−1.19)	−0.7662 (−2.10)
Export/sales	0.0033 (0.80)	0.0075 (1.88)	0.0128 (2.16)	0.0113 (1.99)	0.0185 (2.40)	0.0180 (2.25)
Advertisement/sales	−0.1739 (−6.87)	−0.2117 (−8.62)	−0.0341 (−0.47)	−0.1670 (−2.39)	−0.3942 (−6.01)	−0.5096 (−7.50)
Market share in 4-digit industry	0.1166 (9.53)	0.0956 (8.08)	0.0683 (3.95)	0.0491 (2.96)	0.1071 (5.58)	0.0762 (3.84)
1993 Year dummy	2.9301 (11.65)	3.0677 (12.57)	2.2524 (5.31)	2.2163 (5.46)	3.1592 (16.52)	3.5635 (15.98)
1994 Year dummy	2.9949 (11.99)	3.3263 (13.72)	2.5072 (5.94)	2.3472 (5.81)	3.8991 (17.41)	4.3317 (17.94)
1995 Year dummy	1.9971 (8.08)	2.2535 (9.49)	2.2572 (5.30)	2.2995 (5.64)	2.7729 (5.23)	3.1214 (5.69)
1996 Year dummy	1.0126 (4.15)	1.3196 (5.57)	1.1658 (7.73)	0.9898 (2.42)	1.1422 (2.16)	1.3847 (2.53)

Table 5, part 1: Determinants of firm profitability, Panel A.

6.5 Table 6: Robustness tests

Read:

- Panel A repeats the analysis using ordinary income/sales and net income/sales instead of income/assets.
- Ownership concentration remains positive, and large 70 *chaebol* affiliation remains negative.
- The control–ownership wedge is negative in the sales-based specifications, especially for publicly traded firms.
- Panel B replaces the large 70 *chaebol* classification with the narrower top 30 *chaebol* classification.
- The main results remain: ownership concentration is positive, the wedge is negative, and large group affiliation is negative.

Economic message: The findings are not driven by a specific profitability denominator or by the choice of top 70 rather than top 30 business groups.

Table 6
Robustness tests with different profitability measure and a narrow classification of large *chaebols*
In Panel A, profitability is measured through ordinary income to sales and net income to sales, and size is controlled by the log value of sales. In Panel B, large *chaebol* means the top 30 *chaebols* identified by the government in 1995. The three pairs of columns in each panel show the results data from (a) all sample firms, (b) only publicly traded firms, and (c) firms affiliated with business groups. Controlling for industry and time-fixed effects, within-unit estimation is used. All variables are measured as the deviation from the time-series industry mean of each variable. *T*-statistics are in parentheses.

	All sample firms		Publicly traded firms only		Business group affiliated firms only	
	Ordinary income to sales	Net income to sales	Ordinary income to sales	Net income to sales	Ordinary income to sales	Net income to sales
Log (Sales)	0.1280 (30.89)	0.1586 (31.49)	0.1212 (29.83)	0.3364 (30.55)	0.0292 (13.76)	0.1210 (12.28)
Equity ratio _{it-1}	0.0008 (5.82)	0.0545 (3.85)	0.0008 (5.58)	0.0510 (6.34)	0.0005 (6.08)	0.0469 (5.85)
Ownership concentration	0.0011 (7.82)	0.1094 (7.38)	0.0010 (8.99)	0.0875 (7.81)	0.0014 (2.82)	0.1424 (3.07)
Difference between control rights and ownership rights	−0.0004 (−2.61)	−0.0281 (−1.89)	−0.0006 (−5.26)	−0.0473 (−4.12)	−0.0003 (−0.65)	−0.0248 (−0.62)
Large 70 <i>chaebol</i> dummy	−0.2239 (−14.45)	−23.1761 (−14.40)	−0.2388 (−15.28)	−24.8605 (−15.32)	−0.0538 (−6.74)	−5.3144 (−6.35)

Table 6, part 1: Robustness with sales-based profitability.

	All sample firms		Publicly traded firms only		Business group affiliated firms	
	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets
Log (Asset)	0.4728 (5.64)	0.6163 (7.58)	−0.3219 (−2.25)	−0.1671 (−1.22)	0.4094 (2.92)	0.6794 (4.68)
Equity ratio _{it-1}	0.0717 (31.11)	0.0583 (26.09)	0.0258 (6.95)	0.0222 (6.26)	0.0855 (19.43)	0.0520 (11.43)
Difference between control rights and ownership rights	−0.0014 (−0.56)	0.0003 (0.13)	−0.0204 (−3.96)	−0.0140 (−2.83)	−0.0086 (−1.90)	−0.0438 (−1.03)
Large 70 <i>chaebol</i> Dummy	−3.5456 (−13.01)	−3.3184 (−12.55)	−1.5786 (−3.99)	−1.5616 (−4.13)	−0.6811 (−1.90)	−0.9970 (−2.69)
Export/sales	0.0033 (0.80)	0.0075 (1.87)	0.0122 (2.04)	0.0108 (1.89)	0.0178 (2.30)	0.0173 (2.17)
Advertisement/sales	−0.1720 (−6.78)	−0.2103 (−8.55)	0.0121 (−0.16)	−0.1435 (−2.04)	−0.3794 (−5.77)	−0.4980 (−7.33)
Market share in 4-digit industry	0.1184 (9.67)	0.0974 (8.20)	0.0005 (0.42)	0.0607 (4.62)	0.1117 (5.81)	0.0800 (4.02)
1993 Year dummy	2.8634 (11.56)	3.0180 (12.35)	2.2279 (5.20)	2.2143 (4.40)	3.1332 (5.89)	3.5640 (6.48)
1994 Year dummy	2.9491 (11.78)	3.2917 (13.56)	2.4491 (5.80)	2.3245 (5.70)	3.8975 (7.36)	4.3461 (7.93)
1995 Year dummy	1.9796 (8.00)	2.2425 (9.34)	2.3312 (5.42)	2.3776 (5.78)	2.8269 (5.31)	3.1760 (5.77)
1996 Year dummy	1.0631 (4.10)	1.3139 (5.54)	1.2326 (2.86)	1.0569 (2.56)	1.2066 (2.26)	1.4380 (2.62)
Industry dummies	Included	Included	Included	Included	Included	Included
No. of observations	19,497	19,497	4,702	4,702	5,314	5,314
<i>R</i> ²	0.0900	0.0874	0.1523	0.1351	0.0892	0.0879

Table 5, part 2: Determinants of firm profitability, Panel B.

	All sample firms		Publicly traded firms only		Business group affiliated firms only	
	Ordinary income to sales	Net income to sales	Ordinary income to sales	Net income to sales	Ordinary income to sales	Net income to sales
Log (Asset)	0.8570 (11.15)	0.9320 (12.49)	0.7016 (9.41)	0.7996 (11.06)	0.3796 (3.20)	−0.0448 (−0.37)
Equity ratio _{it-1}	0.0738 (31.99)	0.0603 (26.92)	0.0733 (31.71)	0.0598 (26.67)	0.0277 (7.51)	0.0261 (6.84)
Ownership concentration	0.0196 (8.08)	0.0167 (7.01)	0.0531 (10.40)	0.0480 (9.83)	0.0292 (5.55)	0.0207 (3.79)
Difference between control rights and ownership rights	−0.0049 (−2.02)	−0.0028 (−1.18)	−0.0227 (−4.39)	−0.0155 (−3.13)	−0.0098 (−2.18)	−0.0053 (−1.14)
Large 30 <i>chaebol</i> dummy	−2.8850 (−8.77)	−2.8294 (−8.87)	−3.1313 (−9.42)	−3.0765 (−9.55)	−1.0510 (−2.42)	−0.9741 (−2.35)
Industry dummies	Included	Included	Included	Included	Included	Included
Time dummies	Included	Included	Included	Included	Included	Included
No. of observations	19,497	19,497	19,497	19,497	4,702	4,702
<i>R</i> ²	0.1547	0.1573	0.1520	0.1351	0.3634	0.3510

Table 6, part 2: Robustness using the top 30 chaebol classification.

6.6 Table 7: Time-varying effects of ownership concentration

Read:

- Panel A interacts ownership concentration with the annual GNP growth rate.
- For all sample firms, the interaction is negative, meaning the ownership effect is stronger when growth is lower.
- Panel B allows the ownership slope to vary year by year.
- The 1997 ownership coefficient is the largest in the all-firm regressions.
- Publicly traded firms show high ownership coefficients throughout the period.

Economic message: Governance problems become more costly when the economy weakens. This is consistent with controlling shareholders having stronger incentives to divert resources when distress risk rises.

Table 7
Time-varying effects of ownership concentration
The dependent variables are ordinary income to assets and net income to assets. Panel A reports how the macroeconomic condition affects the effects of ownership concentration on firm profitability. Table B reports the results without the restriction that ownership concentration yields the same slope in each year. The other explanatory variables are firm characteristics. The three pairs of columns in each panel show the results using data from (a) all sample firms, (b) only publicly traded firms, and (c) firms affiliated with business groups. Controlling for industry and time-fixed effects, within-unit estimation is used. All variables are measured as the deviation from the time-series industry mean of each variable. Large 70 *chaebol* is a dummy variable that has a value of 1 when a firm belongs to the largest 70 business groups. *T*-statistics are in parentheses.

	All sample firms		Publicly traded firms only		Business group affiliated firms	
	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets
<i>Panel A:</i>						
Log (Asset)	0.6163 (7.18)	0.7385 (8.87)	0.1345 (0.91)	0.2402 (1.70)	0.6040 (4.27)	0.8168 (5.57)
Equity ratio _{t-1}	0.0721 (31.33)	0.0587 (26.30)	0.0272 (7.40)	0.0235 (6.69)	0.0864 (19.66)	0.0526 (11.56)
Ownership concentration	0.0594 (5.75)	0.0472 (4.70)	0.0687 (3.24)	0.0435 (2.14)	0.0556 (2.48)	0.0482 (2.08)
Ownership concentration * GNP growth rate	-0.0060 (-4.17)	-0.0046 (-3.30)	-0.0025 (-0.88)	0.0004 (0.16)	-0.0037 (-1.23)	-0.0037 (-1.19)
Large 70 <i>chaebol</i> dummy	-3.2642 (-12.15)	-3.0379 (-11.65)	-1.3519 (-3.55)	-1.2599 (-3.45)	-0.4249 (-1.21)	-0.7720 (-2.12)
Export/sales	0.0030 (0.74)	0.0073 (1.82)	0.0127 (2.14)	0.0113 (1.99)	0.0184 (2.39)	0.0178 (2.23)
Advertisement/sales	-0.1740 (-6.87)	-0.2118 (-8.62)	-0.0357 (-0.49)	-0.1667 (-2.39)	-0.3947 (-6.01)	-0.5101 (-7.51)
Market share in 4-digit industry	0.1167 (9.54)	0.0956 (8.06)	0.0683 (3.95)	0.0491 (2.96)	0.1072 (5.59)	0.0762 (3.84)
Industry dummies	Included	Included	Included	Included	Included	Included
Time dummies	Included	Included	Included	Included	Included	Included
No. of observations	19,497	19,497	4,702	4,702	5,314	5,314
<i>R</i> ²	0.1640	0.1463	0.3672	0.3160	0.2877	0.2148

	All sample firms		Publicly traded firms only		Business group affiliated firms	
	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets
<i>Panel B:</i>						
Log (Asset)	0.6197 (7.22)	0.7425 (8.91)	0.1370 (0.93)	0.2441 (1.73)	0.5969 (4.21)	0.8088 (5.52)
Equity ratio _{t-1}	0.0722 (31.37)	0.0588 (26.34)	0.0271 (7.37)	0.0234 (6.64)	0.0864 (19.66)	0.0526 (11.56)
Ownership concentration in 93	0.0194 (3.88)	0.0145 (3.00)	0.0494 (5.66)	0.0376 (4.50)	0.0224 (3.33)	0.0226 (2.24)
Ownership concentration in 94	0.0091 (3.86)	0.0064 (1.36)	0.0443(5.06)	0.0462 (5.51)	0.0294 (3.13)	0.0201 (2.66)
Ownership concentration in 95	0.0094 (2.00)	0.0111 (2.44)	0.0480 (5.05)	0.0449 (4.93)	0.0238 (2.49)	0.0188 (1.90)
Ownership concentration in 96	0.0160 (3.53)	0.0132 (2.99)	0.0581 (5.83)	0.0579 (6.06)	0.0178 (1.86)	0.0089 (0.90)
Ownership concentration in 97	0.0399 (7.30)	0.0352 (6.64)	0.0644 (4.38)	0.0543 (3.87)	0.0573 (3.76)	0.0576 (3.76)
Large 70 <i>chaebol</i> dummy	-3.2777 (-12.20)	-3.0537 (-11.71)	-1.3486 (-3.52)	-1.2576 (-3.43)	-0.4407 (-1.24)	-0.7875 (-2.16)
Export/sales	0.0030 (0.73)	0.0073 (1.82)	0.0126 (2.13)	0.0112 (1.98)	0.0182 (2.36)	0.0176 (2.20)
Advertisement/sales	-0.1725 (-6.81)	-0.2101 (-8.55)	-0.0353 (-0.48)	-0.1664 (-2.38)	-0.3949 (-6.02)	-0.5098 (-7.50)
Market share in 4-digit industry	0.1165 (9.53)	0.0955 (8.05)	0.0680 (3.93)	0.0486 (2.93)	0.1074 (5.60)	0.0764 (3.85)
Industry dummies	Included	Included	Included	Included	Included	Included
Time dummies	Included	Included	Included	Included	Included	Included
No. of observations	19,497	19,497	4,702	4,702	5,314	5,314
<i>R</i> ²	0.1643	0.1467	0.3674	0.3165	0.2883	0.2158

Table 7, part 1: Ownership effects and GNP growth.

Table 7, part 2: Year-specific ownership slopes.

6.7 Table 8: Nonlinearity of ownership effects

Read:

- Panel A estimates piecewise ownership effects over 0–5%, 5–25%, and above 25%.
- Profitability declines at very low ownership levels, rises strongly between 5% and 25%, and rises more gradually above 25%.
- Panel B estimates a cubic ownership relationship.
- The cubic terms indicate a non-monotonic pattern: profitability is generally higher with ownership, but the relation is not purely linear.

Economic message: In Korea's weak governance environment, controlling shareholders could become entrenched even with small ownership stakes. Alignment improves once the controlling family owns a meaningful cash-flow stake.

Table 8

Non-linearity of ownership effects on profitability The dependent variables are ordinary income to assets and net income to assets. Panel A tests for ownership's nonlinear effects using linear, squared, and cubed terms as explanatory variables. Panel B tests for ownership's nonlinear effects using piecewise linear splines from 0, 5%, 5-25%, and 75-100%, as in Morck, Shleifer, and Vishny (1988). The three pairs of columns in each panel show the results using data from (a) all sample firms, (b) only publicly traded firms, and (c) firms affiliated with business groups. Controlling for industry and time-fixed effects, within unit estimation is used. All variables are measured as the deviation from the time-series industry mean of each variable. Large 70 *chaebol* is a dummy variable that has a value of 1 when a firm belongs to the largest 70 business groups. *T*-statistics are in parentheses.

	All sample firms		Publicly traded firms only		Business group affiliated firms only	
	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets
<i>Panel A:</i>						
Log (Assets)	0.6673 (7.75)	0.7934 (9.48)	0.1735 (1.17)	0.2804 (1.98)	0.6260 (4.38)	0.8507 (5.75)
Equity ratio ₋₁	0.0721 (31.39)	0.0587 (26.33)	0.0269 (7.31)	0.0234 (6.65)	0.0866 (19.70)	0.0529 (11.62)
Ownership concentration 0 to 5%	-0.5311 (-5.75)	-0.5522 (-6.16)	-0.3440 (-3.21)	-0.3459 (-3.37)	-0.1529 (-1.11)	-0.2107 (-1.48)
Ownership concentration 5 to 25%	0.1350 (5.55)	0.1332 (5.63)	0.1623 (5.65)	0.1360 (4.94)	0.0748 (1.92)	0.0641 (1.59)
Ownership concentration above 25%	0.0174 (4.88)	0.0156 (4.51)	0.0390 (5.31)	0.0415 (5.90)	0.0266 (3.11)	0.0236 (2.66)
Large 70 <i>chaebol</i> dummy	-3.3195 (-12.16)	-3.1115 (-11.74)	-1.3220 (-3.40)	-1.3098 (-3.52)	-0.4297 (-1.21)	-0.8193 (-2.22)
Export/sales	0.0035 (0.85)	0.0077 (1.93)	0.0131 (2.22)	0.0115 (2.03)	0.0185 (2.39)	0.0178 (2.23)
Advertisement/sales	-0.1772 (-7.00)	-0.2152 (-8.79)	-0.0480 (-0.66)	-0.1758 (-2.52)	-0.3975 (-6.04)	-0.5110 (-7.50)
Market share in 4-digit industry	0.1185 (9.70)	0.0976 (8.23)	0.0678 (3.92)	0.0487 (2.94)	0.0774 (3.89)	0.0774 (3.89)

Table 8, part 1: Piecewise ownership effects.

Industry dummies Time dummies R^2	Included Included 0.1648	Included Included 0.1476	Included Included 0.3694	Included Included 0.3183	Included Included 0.2877	Included Included 0.2151
	All sample firms		Publicly traded firms only		Business group affiliated firms only	
	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets
<i>Panel B:</i>						
Log (Assets)	0.6618 (7.68)	0.7849 (9.39)	0.2193 (1.48)	0.3189 (2.24)	0.6405 (4.48)	0.8587 (5.80)
Equity ratio ₋₁	0.0721 (31.36)	0.0587 (26.30)	0.0269 (7.31)	0.0234 (6.65)	0.0865 (19.69)	0.0528 (11.60)
Ownership concentration	-0.0899 (-4.80)	-0.0857 (-4.80)	-0.0116 (-0.40)	-0.0241 (-1.25)	-0.0489 (-0.67)	-0.0489 (-1.31)
(Ownership concentration) ²	0.0030 (6.14)	0.0030 (6.35)	0.0027 (3.25)	0.0029 (3.73)	0.0020 (1.87)	0.0023 (2.10)
(Ownership concentration) ³	-0.0002 (-6.37)	-0.0002 (-6.53)	-0.0002 (-3.78)	-0.0002 (-4.04)	-0.0002 (-2.03)	-0.0002 (-2.14)
Large 70 <i>chaebol</i> dummy	-3.3072 (-12.11)	-3.1012 (-11.70)	-1.3185 (-3.39)	-1.3056 (-3.51)	-0.3959 (-1.11)	-0.7857 (-2.13)
Export/sales	0.0032 (0.79)	0.0074 (1.86)	0.0128 (2.17)	0.0112 (1.98)	0.0182 (2.36)	0.0175 (2.19)
Advertisement/sales	-0.1755 (-7.15)	-0.2134 (-8.69)	-0.0366 (-0.50)	-0.1657 (-2.38)	-0.4010 (-6.10)	-0.5136 (-7.54)
Market share in 4-digit industry	0.1169 (9.50)	0.0959 (8.09)	0.0630 (3.64)	0.0445 (2.88)	0.1066 (15.50)	0.0761 (3.83)
Industry dummies	Included	Included	Included	Included	Included	Included
Time dummies	Included	Included	Included	Included	Included	Included
No. of observations	19,497	19,497	4,702	4,702	5,314	5,314
R^2	0.1651	0.1477	0.3700	0.3188	0.2881	0.2153

Table 8, part 2: Cubic ownership effects.

6.8 Table 9: Expropriation in publicly traded firms

Read Panel A:

- Investments in affiliated firms are negatively related to profitability.
- Investments in unaffiliated firms are positively related to profitability in the all-firm and public-firm samples.
- The negative affiliated-investment coefficient is especially large for publicly traded firms and business-group firms.

Read Panel B:

- Ownership concentration matters more for public firms, as shown by positive interactions between ownership and the public-firm dummy.
- The interaction between public-firm status and investment in affiliated firms is negative.
- This means moving resources to affiliated firms is especially costly for publicly traded firms.

Economic message: The evidence is consistent with tunneling. Internal capital markets do not simply reallocate funds efficiently; they can move resources away from firms with outside minority shareholders and destroy profitability.

Table 9

Expropriation in publicly traded firms

The dependent variables are ordinary income to assets and net income to assets. Panel A shows that financial investment to affiliated firms affects firm profitability. Panel B shows that controlling shareholder expropriation is more serious in publicly traded firms than privately held firms. The public firm dummy and its interaction terms with financial investment to affiliated firms are included as explanatory variables. Controlling for industry and time-fixed effects, within unit estimation is used. All variables are measured as the deviation from the time-series industry mean of each variable. Large 70 *chaebol* is a dummy variable that has a value of 1 when a firm belongs to one of the largest 70 business groups. *T*-statistics are in parentheses.

	All sample firms		Publicly traded firms only		Business group affiliated firms	
	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets
<i>Panel A:</i>						
Log (Assets)	0.7700 (9.60)	0.8847 (11.48)	0.1975 (0.28)	0.2808 (2.30)	0.7721 (5.92)	0.8716 (6.50)
Equity ratio ₋₁	0.0779 (33.64)	0.0642 (28.65)	0.0287 (7.85)	0.0251 (20.23)	0.0909 (12.38)	0.0568 (12.38)
Ownership concentration	0.0192 (7.93)	0.0168 (7.17)	0.0488 (9.88)	0.0443 (9.38)	0.0297 (5.70)	0.0216 (4.05)
Large <i>chaebol</i> dummy	-3.1891 (-11.89)	-2.8755 (-11.08)	-1.4610 (-3.94)	-1.3195 (-3.73)	-0.6687 (-1.91)	-0.9485 (-2.66)
Investment/assets in affiliated firms	-0.0367 (-2.99)	-0.0261 (-2.20)	-0.0599 (0.980)	-0.0041 (0.076)	-0.0443 (0.061)	-0.0443 (0.044)
Investment/assets in unaffiliated firms	0.0541 (5.42)	0.0560 (5.81)	0.0560 (5.22)	0.0560 (5.22)	0.0560 (5.22)	0.0560 (5.22)
Export/sales	0.0033 (0.79)	0.0077 (1.94)	0.0138 (2.37)	0.0138 (2.37)	0.0138 (2.37)	0.0138 (2.37)
Advertisement/sales	-0.1664 (-6.43)	-0.1922 (-7.96)	-0.0142 (-0.22)	-0.1190 (-1.54)	-0.3567 (-5.45)	-0.4411 (-6.99)
Market share in 4-digit industry	0.0576 (4.07)	0.0381 (4.44)	0.0406 (3.15)	0.0322 (2.61)	0.0663 (4.59)	0.0576 (3.91)
Industry dummies	Included	Included	Included	Included	Included	Included
Time dummies	Included	Included	Included	Included	Included	Included
R^2	0.1245	0.1122	0.3542	0.3043	0.2120	0.1556

Table 9, part 1: Affiliated-firm investment and profitability.

	All sample firms		Business group affiliated firms	
	Ordinary income to assets	Net income to assets	Ordinary income to assets	Net income to assets
<i>Panel B:</i>				
Log (Assets)	0.9574 (10.48)	1.0119 (11.45)	1.0139 (6.66)	1.0489 (6.74)
Equity ratio ₋₁	0.0782 (33.74)	0.0643 (28.71)	0.0908 (20.11)	0.0572 (12.39)
Ownership concentration	0.0120 (4.66)	0.0103 (4.14)	0.0237 (3.92)	0.0151 (2.44)
Ownership concentration *Public firm dummy	0.0473 (7.69)	0.0434 (7.30)	0.0197 (1.91)	0.0202 (1.93)
Public firm dummy	-2.2156 (-5.04)	-1.4266 (-3.35)	-2.7186 (-4.03)	-1.2477 (-1.80)
Large <i>chaebol</i> dummy	-3.0742 (-11.42)	-2.7536 (-10.58)	-0.8228 (-2.31)	-1.0599 (-2.89)
Investment/assets in affiliated firms	-0.0154 (-1.14)	-0.0040 (-0.30)	-0.0521 (-0.53)	-0.0152 (-0.53)
Investment/assets in unaffiliated firms	0.0460 (4.20)	0.0524 (4.95)	-0.0339 (-1.40)	0.0349 (1.41)
Export/sales	0.0046 (0.97)	0.0085 (2.14)	0.0186 (2.39)	0.0198 (2.49)
Advertisement/sales	-0.1632 (-6.55)	-0.1942 (-8.06)	-0.3436 (-5.56)	-0.4450 (-7.05)
Market share in 4-digit industry	0.0572 (7.03)	0.0381 (4.34)	0.0651 (4.51)	0.0576 (3.90)
Investment/assets in affiliated firms *public firm dummy	-0.0762 (-2.45)	-0.0928 (-3.09)	-0.0334 (-0.67)	-0.0818 (-1.61)
Investment/assets in unaffiliated firms	0.0344 (1.35)	0.0344 (1.35)	0.0284 (1.15)	0.0348 (0.83)
Industry dummies	Included	Included	Included	Included
Time dummies	Included	Included	Included	Included
No. of observations	19,497	19,497	5,314	5,314
R^2	0.1278	0.1153	0.2151	0.1570

Table 9, part 2: Public-firm interactions and expropriation.

6.9 Table 10: List of top 30 business groups

Read:

- Table 10 lists the top 30 Korean business groups from 1993 to 1997.
- Hyundai, Samsung, Daewoo, LG, and SK are consistently among the largest groups.
- The table provides the appendix classification used to define the large *chaebol* group in robustness tests.

Economic message: The business-group classification is not abstract. The governance effects are tied to Korea's largest and most economically important conglomerates.

Table 10

(as of the end of the year)

Ranking	1993	1994	1995	1996	1997
1	Hyundai	Hyundai	Hyundai	Hyundai	Hyundai
2	Samsung	Daewoo	Samsung	Samsung	Samsung
3	Daewoo	Samsung	LG	LG	Daewoo
4	LG	LG	Daewoo	Daewoo	LG
5	SK	SK	SK	SK	SK
6	Hanjin	Hanjin	Ssangyong	Ssangyong	Hanjin
7	Ssangyong	Ssangyong	Hanjin	Hanjin	Ssangyong
8	Kia	Kia	Kia	Kia	Hanwha
9	Hanwha	Hanwha	Hanwha	Hanwha	Kumho
10	Lotte	Lotte	Lotte	Lotte	Dongah
11	Kumho	Kumho	Kumho	Kumho	Lotte
12	Daelim	Daelim	Doosan	Halla	Halla
13	Doosan	Doosan	Daelim	Dongah	Daelim
14	Dongah	Dongah	Hanbo	Doosan	Doosan
15	Hanil	Hyosung	Dongah	Daelim	Hansol
16	Hyosung	Hanil	Halla	Hansol	Hyosung
17	Dongkuk	Halla	Hyosung	Hyosung	Kohab
18	Sammi	Dongkuk	Dongkuk	Dongkuk	Kolon
19	Halla	Sammi	Jinro	Jinro	Dongkuk
20	Hanyang	Tongyang	Kolon	Kolon	Dongbu
21	Tongyang	Kolon	Tongyang	Kohab	Anam
22	Kolon	Jinro	Hansol	Dongbu	Jinro
23	Jinro	Kohab	Dongbu	Tongyang	Tongyang
24	Dongbu	Woosung	Kohab	Haitai	Haitai
25	Kohab	Dongbu	Haitai	New core	Shinho
26	Kukdong	Haitai	Sammi	Anam	Daesang
27	Woosung	Kukdong	Hanil	Hanil	New Core
28	Haitai	Hanbo	Kukdong	Keopyong	Keopyong
29	Byuksan	Daesang	New Core	Daesang	Kangwon
30	Daesang	Byuksan	Byuksan	Shinho	Saehan

Note: When a group changes its names, the latest name has been used.

Table 10: Top 30 Korean business groups, 1993–1997.

7 Short summary

- The paper studies whether weak corporate governance reduced Korean firm profitability before the 1997 crisis.
- The sample contains 5,829 firms and 19,497 firm-year observations during 1993–1997.
- The paper uses accounting profitability because it applies to both public and private firms and is closely related to financial survivability.
- Controlling-family ownership is positively related to profitability.
- Firms with larger control–ownership disparities have lower profitability, especially publicly traded firms and business-group firms.

- Large *chaebol*-affiliated firms underperform independent firms.
- The ownership effect is strongest in 1997 and stronger when aggregate growth is lower.
- The ownership-profitability relation is nonlinear, indicating both entrenchment and alignment forces.
- Investments in affiliated firms reduce profitability, while investments in unaffiliated firms tend to increase profitability.
- The negative effect of affiliated-firm investment is stronger in public firms, consistent with tunneling from firms with outside minority shareholders.
- The results suggest that poor governance weakened firm fundamentals before the crisis, rather than merely worsening crisis-period stock returns.
- The paper links corporate governance to macro-financial fragility through persistent low profitability and inefficient capital allocation.

8 Conclusion

8.1 Core conclusion

Joh shows that weak corporate governance lowered Korean firm profitability before the 1997 crisis. Firms with low controlling-family ownership, large control–ownership wedges, large *chaebol* affiliation, and high investment in affiliated firms performed worse.

8.2 Economic intuition

A controlling shareholder who has effective control but limited cash-flow ownership does not fully internalize the cost of inefficient investment or resource diversion. In business groups, pyramidal ownership and interlocking shareholdings give the controlling family more tools to transfer resources across affiliates. These transfers can protect the controlling family while reducing profitability at individual firms.

8.3 Incremental contribution revisited

The paper’s key contribution is to show that governance problems were visible before the crisis. It moves the discussion from crisis-period market-value collapses to pre-crisis operating performance, and it provides direct evidence that business-group internal capital markets could destroy value.

8.4 Implications

The findings imply that financial-stability analysis should examine corporate governance and corporate profitability before crises. High leverage is dangerous when governance allows unprofitable firms to survive, invest inefficiently, and transfer resources away from minority shareholders.

8.5 Limitations

The empirical design is observational. Ownership and investment decisions can be endogenous, and accounting profitability can contain reporting noise. The evidence is from Korea during a specific pre-crisis period, so external validity should be assessed carefully. The affiliated-investment regressions are especially suggestive rather than fully causal because firms choose where to invest their financial assets.

8.6 Takeaway

The enduring takeaway is that corporate governance matters before crises. Weak investor protection and controlling-shareholder expropriation can reduce profitability for years, weaken balance sheets, generate nonperforming loans, and make the entire financial system more fragile.